Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/30/2021
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Project Title:	Vestibular Modulation of Autonomic Responses to Orthostasis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-03
Start Date:	11/15/2000	**End Date:**	04/01/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG9-1262		
Performance Goal No.:			
Performance Goal Text:			
	Orthostatic intolerance and related functional impairment remain a common problem for astronauts upon return from space. Despite considerable effort to develop countermeasures for this problem, a fundamental question remains unanswered: What is the cause of post-flight orthostatic intolerance? The cause of syncope is also a common and difficult problem to diagnose in the general population. Post-flight symptoms vary widely and include both vestibular and orthostatic intolerance manifestations. Although it is known that vestibular signals project to the vasomotor centers of the brainstem in cats, there has been no prospective study of vestibular system effects on blood pressure control during orthostasis or on orthostatic tolerance in humans. Most fundamental, it remains unknown whether otolith or semicircular canal inputs play an important role in the cardiovascular adjustments to the upright position. Therefore, this proposal will address the following aims: to determine the role of otolith input in mediating the cardiovascular adjustments to the upright position; to determine the effect of 60o how vestibular stimulation affects the occurrence of pre-syncopal symptoms and vasovagal reaction during an orthostatic stress;3) to determine the effect of semicircular		

Task Description:	canal stimulation (with caloric testing) on the stability of blood pressure regulation during orthostasis and orthostatic tolerance; 4) to determine the effect of promethazine treatment on the stability of blood pressure regulation and orthostatic tolerance during unperturbed orthostasis or orthostasis with accompanying vestibular stimulation. Four studies will address individually each specific aim. Study 1 will address aim 1 by assessing the cardiovascular and autonomic responses to graded levels of orthostasis (head-up tilt) with and without graded levels of otolith orientation (to gravity). Study 2 will address aim 2 by assessing the cardiovascular and autonomic responses to sustained orthostasis with and without otolith input. Study 3 will address aim 3 by assessing the cardiovascular and autonomic responses to sustained orthostasis with and without semicircular canal stimulation (caloric testing). Study 4 will address aim 4 by assessing orthostatic responses and tolerance with and without pre-treatment with promethazine.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Direct impact on human function on Earth is not defined. However, the preliminary results suggest that in normal healthy individuals that the otolith system does not adversely affect the autonomic response to the upright position, and thus, play an important role in the cause of syncope. A direct link to syncope remains to be addressed in ongoing studies.
Task Progress:	[Ed. note: report compiled from information received June 2021; report is dated April 2005] OBJECTIVES: Orthostatic intolerance and related functional impairment remain a common problem for astronauts upon return from space. Despite considerable effort to develop countermeasures for this problem, a fundamental question remains unanswered: What is the cause of orthostatic intolerance? Post flight symptoms vary widely and include both vestibular and orthostatic intolerance manifestations. The role of vestibular disturbances in orthostatic intolerance is unknown. Although it is known that vestibular signals project to the vasomotor centers of the brainstem in cats, there has been no prospective study of vestibular system effects on blood pressure control during orthostasis or on orthostatic tolerance in humans. Most fundamental, it remains unknown whether otolith inputs play an important role in the cardiovascular adjustments to the upright position in humans. The altered vestibular function after space flight may influence orthostatic tolerance indirectly (producing intolerable symptoms requiring recumbent posture) and/or directly (by affecting autonomic responses to orthostasis). Post flight vestibular disturbances are complex and probably include some involvement of otolith inputs, semicircular canal inputs, and related inputs. The purpose of this study was three-fold: 1) to determine the otolith contribution to the cardiovascular adjustments to the upright position; 2) to determine whether altering otolith or semicircular canal inputs affect blood pressure regulation during orthostasis; 3) to determine whether altering otolith or semicircular canal inputs affect orthostatic tolerance. The specific aims of the study were as follows: 1. To determine the role of otolith input in mediating the cardiovascular adjustments to the upright position. 2. To determine how otolith orientation affects the occurrence of pre-syncopal symptoms during an orthostatic stress. 3. To determine the effect of semicircular canal stimulation (with caloric testing) on the cardiovascular adjustments to orthostasis. 4. To determine the effect of semicircular canal stimulation (with caloric testing) on orthostatic tolerance. APPROACH: Study 1: This study was performed to address specific aim 1. The response to seven perturbations involving combinations of head-up tilt (0, 30, and 60 degree) and neck position to affect otolith orientation. (30 degree flexion or extension) were obtained in random order. Study 2: This study addressed specific aim 2 by assessing the heart rate, arterial pressure, sympathetic nerve activity, heart rate, and blood pressure variances, respiration, responses to sustained orthostasis (30 degree head-up tilt plus lower body negative pressure at 10, 20, and 30 mmHg) with and without otolith input (0 degrees versus 60 degrees otolith orientation). Study 3: This study addressed specific aims 3 and 4 by assessing the heart rate, arterial pressure, SNA, heart rate, and blood pressure variances, responses to graded orthostasis (lower body negative pressure at -30, -40, -50, and -60 mmHg) with and without semicircular canal stimulation (caloric ear canal irrigation). RESULTS: Study 1: Within the first 20 seconds of tilt, arterial pressure increased most obviously in the 0 to 1 Gz otolith condition. Thoracic impedance tended to increase more in otolith-constant conditions. But no dependent variable differed significantly between tilt conditions, and no significant time x tilt interactions emerged. Study 2: There was a trend for greater increases in heart rate, but not for differences in arterial pressure or vascular resistance during the graded orthostasis. In addition, changes in low frequency and high frequency spectral power trended toward significance, but did not achieve significance. In some subjects, this effect suggests that autonomic control was modestly affected by the otolith stimulus during graded orthostasis. Study 3: Significant nystagmus was provoked by the caloric stimulus in each subject with insignificant effects on baseline heart rate, mean arterial pressure, or sympathetic nerve activity consistent with previous pilot data and previous findings. The cold caloric stimulation performed during LBNP (lower body negative pressure) resulted in similar changes as the thermoneutral stimulus in heart rate and blood pressure suggesting that there was not a significant effect of semicircular canal stimulation.
Bibliography Type:	Description: (Last Updated: 02/27/2004)